1. OSNOVNI SIGNALI

ZVONKO

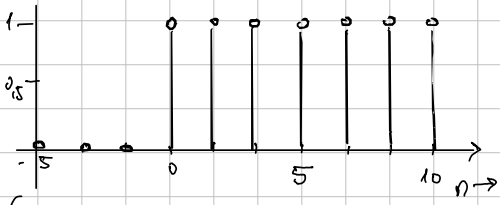
Jedinični impuls

Vremenski kontinuirani skok - vremenska jedinična step funkcija

▸ vremenski diskretan jed. skok u μ

$\forall n \in \mathbb{Z}$

$$\mu[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$$



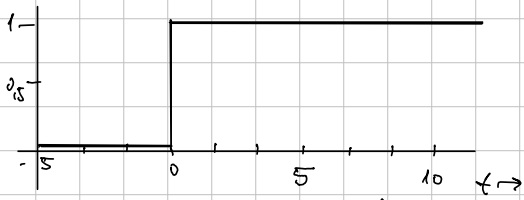
→ dobro opisuje signali koji brzo odjednom (npr. sklopka)

domena mu je $\mathbb{Z}$

→ diskretni step → u 0 je definiran

▸ vremenski kontinuirani jed. skok u μ

$$\mu(t) = \begin{cases} 1, & t < 0 \\ 0, & t > 0 \end{cases} \quad \forall t \in \mathbb{R}$$



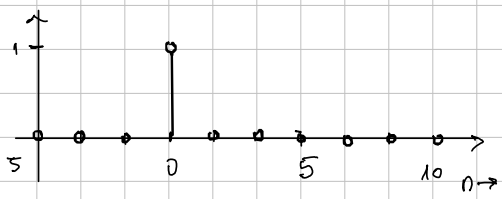
domena je $\mathbb{R}$

▸ Vremenski diskretan jedinični impuls - Kroneckerova delta funkcija

. vremenski diskretan jed. imp δ je vremenski diskretan signal:

$\forall n \in \mathbb{Z}$

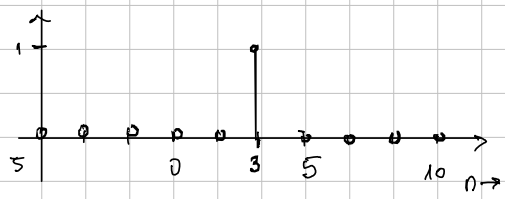
$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$



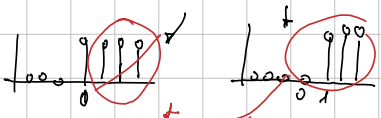
. za m koraka pomaknuti vremenski diskretan jed. imp.

$\forall n \in \mathbb{Z}$, $m \in \mathbb{Z}$

$$\delta[n-m] = \begin{cases} 1, & n = m \\ 0, & n \neq m \end{cases}$$



=> * jedinični Kronecker δ => $\delta[1] = \mu[1] - \mu[n-1]$ (Kronecker = Dirac)



oduzmemo sve osim tj 1 => Kronecker

=> kako definirati step?

$$\boxed{\mu[n] = \sum_{m=0}^{\infty} \delta[n-m]}$$

→ primjer $\mu[-3] = 0$

$$\mu[-3] = \underbrace{\delta[-3-0]}_{=0} + \underbrace{\delta[-3-1]}_{=0} + \dots = 0 \checkmark$$

primjer $\mu[0] = 1$

$$\sum_{m=0}^{\infty} \delta[0-m] = \underbrace{\delta[0-0]}_{1} + \underbrace{\delta[0-1]}_{=0} + \underbrace{\delta[0-2]}_{=0} \dots = \underline{\underline{1}} \checkmark$$

$$\mu[n] = \sum_{m=0}^{\infty} \delta[n-m]$$, uvedemo zamjenu var: $k = n-m$

$\Rightarrow m = n-k$

$$\Rightarrow \mu[n] = \sum_{k=n}^{-\infty} \delta[k] = \sum_{k=-\infty}^{n} \delta[k]$$

(od $n \to -\infty$) (od $-\infty \to n$) $\to$ isto je

* za razliku od integrala, zamjena smjera ne mijenjamo predznak

Osnovna svojstva jediničnog impulsa:

Veza jediničnog skoka i jediničnog imp.:

Za vremenski disk. skok: $\forall n \in \mathbb{Z}$, $\mu[n] = \sum_{m=-\infty}^{n} \delta[m]$

a za vremenski disk jed imp: $\forall n \in \mathbb{Z}$ $\delta[n] = \mu[n] - \mu[n-1]$

Svojstvo očitavanja jed imp: $f[n]\,\delta[n-n_0] = f[n_0]\,\delta[n-n_0]$

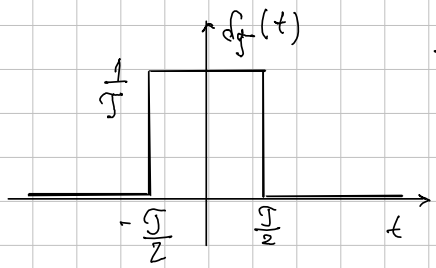
Diracova delta funkcija – vremenski kont. jed. imp

$\Rightarrow$ definiramo kao: $\delta(t) = 0$ za $t \neq 0$ $\to \int_{-\infty}^{\infty} \delta(t)\,dt = 1$, $\forall t \in \mathbb{R}$

pri čemu za pomak impulsa vrijedi: $\delta(t-t_0) = 0$, za $t \neq t_0$

$$\int_{-\infty}^{\infty} \delta(t-t_0)\,dt = 1$$

- izvod za Diracovu δ fju započinje s def. pravokutnog impulsa površine = 1



$$\forall t \in \mathbb{R} \to \delta_\tau(t) = \begin{cases} \frac{1}{\tau}, & -\frac{\tau}{2} < t < \frac{\tau}{2} \\ 0, & t < \frac{-\tau}{2};\ t \geq \frac{\tau}{2} \end{cases}$$

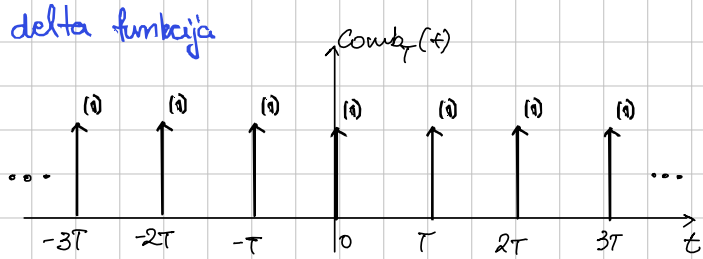
Gaussova krivulja isto ima uvijek površinu 1 $\Rightarrow$ smanjivanjem σ dobivamo Diraca

Svojstva Diraca

- $\int_{-\varepsilon}^{\varepsilon} \delta(t)\,dt = 1$, $\forall \varepsilon > 0$
- $\int_{-\infty}^{\infty} f(t)\delta(t)\,dt = f(0)$, ako je f neprekinuta u 0
- $f(t)\,\delta(t-t_0) = f(t_0)\,\delta(t-t_0)$, ako je f neprekinuta u t_0
- Homogenost: $\delta(at) = \frac{1}{|a|}\delta(t)$
- Parnost: $\delta(-t) = \delta(t)$
- Konvolucija $f * \delta = f$ tj. $\int_{-\infty}^{\infty} f(\tau)\delta(t-\tau)\,d\tau = f(t)$, ako je f neprekinuta fja

Niz Diracovih delta funkcija

→ $comb_T$ = češalj

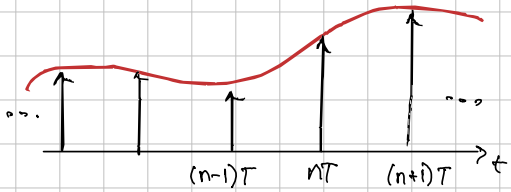


$\forall t \in \mathbb{R}$

$$comb_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - mT), \quad m \in \mathbb{Z}, \; T \in \mathbb{R}$$

Produkt niza Diracovih δ fja i vremenski kont signala:

$t = nT, \; n \in \mathbb{Z}, \; T \in \mathbb{R}$ $\quad f_\delta(t) = f(t) \sum_{n=-\infty}^{\infty} \delta(t - nT) = \sum_{n=-\infty}^{\infty} f(nT)\,\delta(t - nT)$



→ $f(t)\, comb_T(t)$

Nastavak svojstava Diracove δ fje

Deriv. Dirac δ fje N-ta deriv Dirac. δ fje $\delta^{(N)}$ def:

$$\int_{-\infty}^{\infty} f(t)\, \delta^{(N)}(t)\, dt = (-1)^N \cdot f^{(N)}(0)$$

$* \int_{-\infty}^{\infty} f(t)\, \delta(t)\, dt = f(0)$ → radi vrijednost fje u nuli

Veza jediničnog impulsa i jed skoka

$\forall t \in \mathbb{R}, \quad \delta(t) = \dfrac{d\mu(t)}{dt} = \mu'(t)$

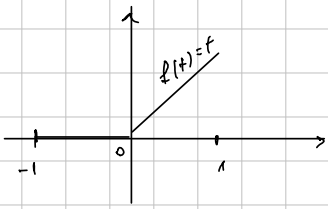
$$\lim_{T \to \infty} \int_{-T}^{T} f(t) \cdot \mu'\, dt = t \cdot \mu \Big|_{-T}^{T} - \int_{-T}^{T} f' \mu(t)\, dt = f(T) - \int_0^T f' dt$$

$$= f(T) - f\Big|_0^T = f(T) - f(T) + f(0) \Rightarrow \lim_{T \to \infty} \int_{-T}^{T} f(t) \cdot \frac{d\mu(t)}{dt} = f(0)$$

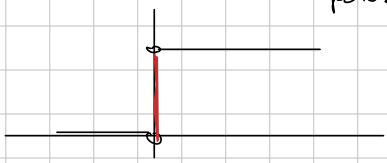
$\forall t \in \mathbb{R}, \quad \mu(t) = \int_{-\infty}^{t} \delta(\lambda)\, d\lambda$

$t < 0, \quad \int_{-\infty}^{t} \delta(\lambda)\, d\lambda = \emptyset \quad ; \quad t > 0, \; \int_{-\infty}^{t} \delta(\lambda)\, d\lambda = 1$ jer skidamo tu vrijednost 1 u 0

Generalizirana derivacija je hibridna fija da kada je integriramo dobijemo polaznu fiju



deriv.
$\Longrightarrow$

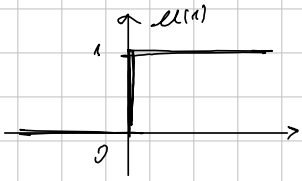


Znači ako gledamo generalno, iako polazna fija nema deriv u svakoj tački → ima derivaciju ji integracijom imamo tu polaznu

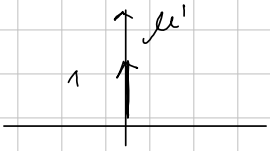
Generalizirana derivacija kontinuirane fije: (oslabljeni pojam derivacije)

$t = t_0$

$$\longrightarrow \quad \forall t \in \mathbb{R} \quad \frac{d}{dt}(g(t)) = \frac{d}{dt}(g(t))_{t \neq t_0} + \left[g(t_0^+) - g(t_0^-)\right]\delta(t - t_0)$$

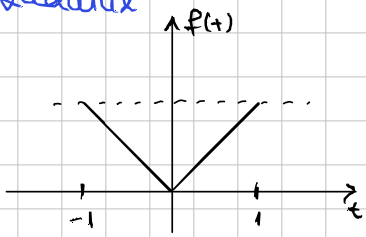


$$\mu(t) = \begin{cases} 0, & t \in \mathbb{R} \setminus \{0\} \end{cases}$$

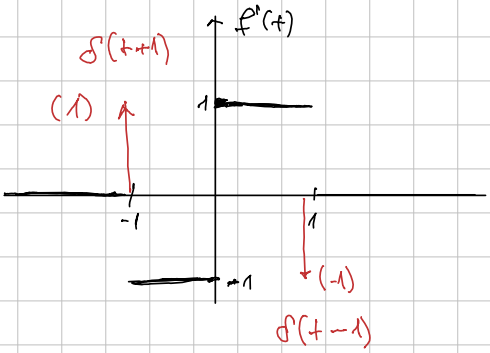


$$\mu' = 0 + (1-0)\,\delta(t)$$

Zadatak:



$-t \to -1 \qquad t \to 1$



Drugi način:

$$f(t) = -t\left(\mu(t) - \mu(t+1)\right) + t\left(\mu(t-1) - \mu(t)\right)$$

$$f(t) = -t\mu(t) + t\mu(t+1) + t\mu(t-1) - t\mu(t)$$

$$f(t) = -2t\mu(t) + t\mu(t+1) + t\mu(t-1)$$

$$f' = -2\mu(t) - 2t\underbrace{\delta(t)}_{=0} + \mu(t+1) + \underbrace{t}_{=-1}\delta(t+1) + \mu(t-1) + \underbrace{t}_{=1}\delta(t-1)$$

svojstvo: $f(t)\,\delta(t - t_0) = f(t_0)\cdot\delta(t - t_0)$

$$f' = -2\mu(t) + \mu(t+1) - \delta(t+1) + \mu(t-1) - \delta(t-1)$$

Vremenski kontinuirani eksponenc. signal

$f: \mathbb{R} \to \mathbb{C}$ $\quad s_0 = \sigma_0 + j\omega_0 \in \mathbb{C}$

$* \cos \omega_0 t = \dfrac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}$

$C = Ae^{j\varphi} \in \mathbb{C}$ $\quad A, \varphi \in \mathbb{R}$

$\sin \omega_0 t = \dfrac{e^{j\omega_0 t} - e^{-j\omega_0 t}}{2}$

$f(t) = Ce^{st} = Ae^{j\varphi} e^{(\sigma_0 + j\omega_0)t} = Ae^{\sigma_0 t} e^{j(\omega_0 t + \varphi)}$

modul $f(t) \to |f(t)| = |Ae^{\sigma_0 t} e^{j(\omega_0 t + \varphi)}| = |Ae^{\sigma_0 t}| \, |e^{j(\omega_0 t + \varphi)}|$

$|e^{j\varphi}| = 1$ uvijek $\to |e^{j(\omega_0 t + \varphi)}| = 1$ $\quad \hookrightarrow |f(t)| = |A| e^{\sigma_0 t}$

Vremenski diskretan kompleksni eksponencijalni signal

$f: \mathbb{Z} \to \mathbb{C}$; $f[n] = Cg^n$ gdje su $C, g \in \mathbb{C}$

$f[n] = C \cdot e^{s_0 n} = C \cdot \underbrace{(e^{s_0})}_{g}{}^n$ $\quad f(t) \to C \cdot g^n$

// opet neki kompleksni jer je $s_0 \in \mathbb{C}$

Sinusoidni signal → vremenski diskretan sinusoidni signal

$f: \mathbb{Z} \to \mathbb{R}$ $\quad f[n] = A\cos(\Omega_0 n + \theta)$

* Svaki diskretan sinusoidni signal nije nužno periodičan

$f[n] = \cos[\Omega_0 n]$

$f[n+N] = \cos[\Omega_0 (n+N)] = f[n]$ $\quad \exists N \in \mathbb{N}$ → mora vrijediti ako je diskr. signal periodičan

$\cos[\Omega_0 (n+N)] = \cos \Omega_0 n$

$\cos[\Omega_0 n + \Omega_0 N] = \cos \Omega_0 n$ $\quad$ za koji $N \in \mathbb{N}$?

$\Omega_0 N = 2\pi k$, $k \in \mathbb{Z}$ (periodičnost kosinusa je za $2\pi k$)

$N = \dfrac{2\pi}{\Omega_0} k \in \mathbb{N}$? (kada je to prirodan br.?)

$\Omega_0 = \dfrac{m}{l} \cdot \pi$; do kraja skraćen razlomak $\dfrac{m}{l}$ → $m \in \mathbb{Z}$, $l \in \mathbb{N}$

kakav je k?

$N = \dfrac{2\pi}{\frac{m\pi}{l}} k = \dfrac{2l}{m} \cdot k = N$ $\quad \Rightarrow N_0 = \begin{cases} 2l, & m = \text{neparan} \quad (k = m) \\ l, & m = \text{paran} \quad (k = \frac{m}{2}) \end{cases}$

Zadatak: $f[n] = \cos \frac{4\pi}{3} n$ Koliki mu je osnovni signal? Je li paran ili neparan?

$\Omega_0 = \frac{4\pi}{3}$ $m = 4 \rightarrow$ paran $\rightarrow k = \frac{m}{2}$

$\ell = 3$

$$N_0 = \frac{2\ell}{m} \cdot k = \frac{6}{\cancel{4}_2} \cdot \cancel{2} \Rightarrow \underline{\underline{N_0 = 3}}$$

(1) Disk. signali nisu nužno periodični => samo kada su racionalan broj × k

(2) $f_1[n] = \cos[\pi n]$ } oba su periodična s periodom 2

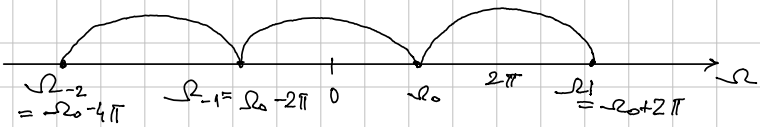
$f_2[n] = \cos[3\pi n] = \cos[\pi n + 2\pi n] = \cos[\pi n] = f_1(n)$ ✓

↳ to su dva različita zapisa istog signala

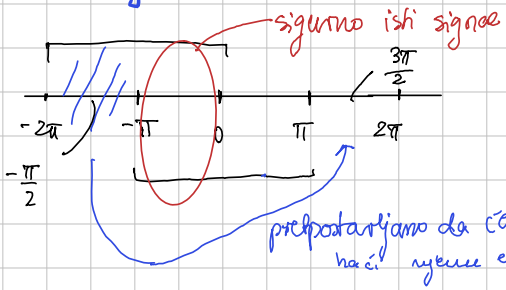
$f_0[n] = \cos(\Omega_0 n)$

$f_k[n] = \cos(\Omega_k n)$ → $\Omega_k = \Omega_0 + 2k\pi, \; k \in \mathbb{Z}$

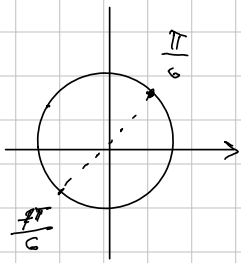
$$f_k[n] \Rightarrow \cos((\Omega_0 + 2k\pi)n) = \cos(\Omega_0 n + 2k\pi \cdot n) = \cos(\Omega_0 n) = f_0[n] \checkmark$$



(3) Dovoljno je razmatrati samo vremenski diskretne sinusoide čiji su frekvencije unutar intervala $0 \leq \Omega \leq 2\pi$ ili $-\pi < \Omega \leq \pi$



→ kao kružnica



$$\cos\left(-\frac{\pi}{2} n\right) = \cos\left(\frac{3\pi}{2} n\right)$$